

Numerical Series

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Numerical Series

Introduction: Numerical Series

- ▶ We introduce **numerical series**, which extend the operation of addition to an **infinite number of terms**
- ▶ A series is studied through the **sequence of partial sums**: its limit determines the **behavior** of the series
- ▶ We will see the definitions of series, partial sums, behavior and sum of a series, with a first example of a divergent series

- ▶ We now introduce numerical series, which extend the operation of addition to an infinite number of terms.

The sum of infinitely many terms, even if they are all positive, can give a finite result.

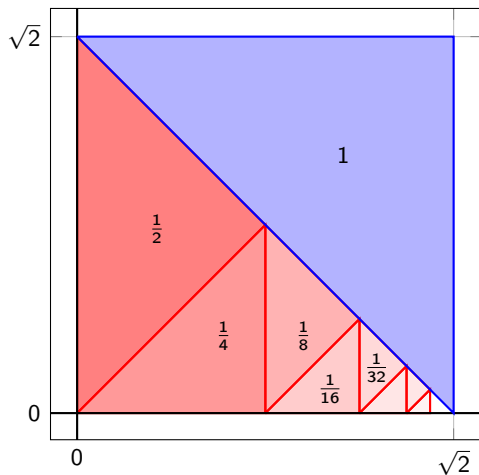
A Geometric Example

- Imagine measuring a square of area 2 using the following procedure: we divide the square in half along the diagonal and measure the first right triangle: we obtain 1; then we divide the second right triangle in half and measure the first of the two resulting right triangles: we obtain $\frac{1}{2}$; the remaining right triangle is again divided in half. . . and so on indefinitely.
- We obtain the infinite sum:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots + \frac{1}{2^k} + \dots = \sum_{k=0}^{\infty} \frac{1}{2^k}$$

which, by the way it was constructed, must give 2 as its result.

A Geometric Example



Definition 1: of numerical series

Given a sequence $\{a_k\}_{k \in \mathbb{N}}$, we call **numerical series** of the terms a_k the expression:

$$\sum_{k=0}^{\infty} a_k$$

- It is read “series (or also sum) for k from 0 to $+\infty$ of a_k ”. The values a_k are called the **general terms of the series**.

Definition 2: of sequence of partial sums

The numbers

$$s_n = \sum_{k=0}^n a_k = a_0 + a_1 + \dots + a_n, \quad \forall n \in \mathbb{N}$$

*are called the **n-th partial sums** of the series and they define the sequence $\{s_n\}$ of partial sums.*

Behavior of a Series

- ▶ The **behavior (character)** of the series is determined by the limit of the sequence $\{s_n\}$ as n tends to infinity.
- ▶ We say that a series is **convergent**, **divergent**, **irregular**, if the **sequence** $\{s_n\}$ of partial sums is **convergent**, **divergent** or **irregular**, respectively.

Sum of a Series

Definition 3: of sum of the series

If the sequence $\{s_n\}$ of partial sums is **convergent**, that is, if

$$s_n \rightarrow s \in \mathbb{R} \quad \text{as } n \rightarrow +\infty$$

we say that s is **the sum of the series**, and we write $\sum_{k=0}^{\infty} a_k = s$.

Sum of a Series

- ▶ Hence, if the sequence $\{s_n\}$ is **convergent**, we have:

$$\sum_{k=0}^{\infty} a_k = \lim_{n \rightarrow +\infty} \sum_{k=0}^n a_k = \lim_{n \rightarrow +\infty} s_n = s$$

- ▶ The series makes precise the idea of a sum of infinitely many terms, that is, we compute the limit as $n \rightarrow +\infty$ of the finite sum of the first n terms.
- ▶ If, instead of summing starting from 0, we start from an index $n_0 > 0$, we write $\sum_{k=n_0}^{\infty} a_k$
- ▶ To denote a generic numerical series without specifying the starting index n_0 we will use the symbol $\sum a_k$

Two Sequences

A generic numerical series $\sum a_k$ always involves **two different sequences**:

1. the sequence $\{a_k\}$ of the **terms of the series**
2. the sequence $\{s_n\}$ of **partial sums**

Example (1/2)

Let us determine the behavior of the series:

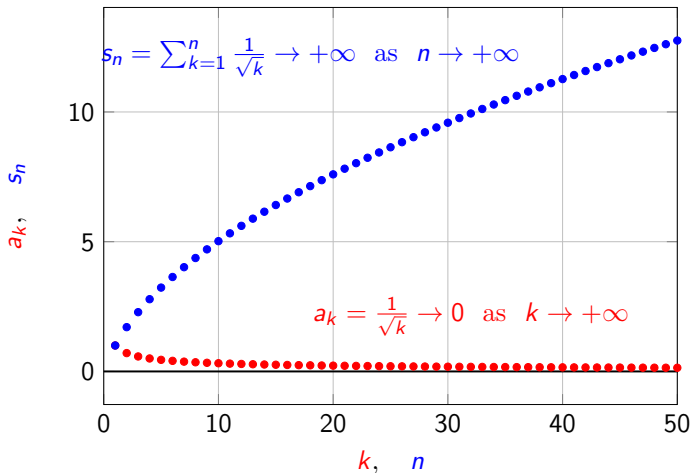
$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$$

We have:

$$\begin{aligned} s_n &= \sum_{k=1}^n \frac{1}{\sqrt{k}} = \underbrace{1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}}_{n \text{ terms, each } \geq \frac{1}{\sqrt{n}}} \\ &\geq n \cdot \frac{1}{\sqrt{n}} = \sqrt{n} \rightarrow +\infty \text{ as } n \rightarrow +\infty \end{aligned}$$

Hence the sequence of partial sums $\{s_n\}$ is divergent by the comparison theorem for sequences and, consequently, the series is divergent.

Example (2/2)



Summary

Numerical series:

- ▶ A series $\sum_{k=0}^{\infty} a_k$ is a “sum of infinitely many terms” a_k
- ▶ The partial sums $s_n = a_0 + \cdots + a_n$ form the sequence $\{s_n\}$
- ▶ The **behavior** of the series (convergent, divergent, irregular) is that of $\{s_n\}$
- ▶ If $s_n \rightarrow s \in \mathbb{R}$, s is the **sum** of the series

Example:

- ▶ $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ diverges, since $s_n \geq \sqrt{n}$

Main Properties of Numerical Series



Introduction: Main Properties of Numerical Series

- ▶ We see that series with **non-negative** terms cannot be irregular: they either converge or diverge
- ▶ We prove the **necessary condition** for convergence: if $\sum a_k$ converges then $a_k \rightarrow 0$
- ▶ This condition is **not sufficient**, but it is a quick tool to recognize series that do not converge

Series with Non-negative Terms

Observation 1

If a sequence $\{a_k\}$ has non-negative terms, that is, $a_k \geq 0$, $\forall k$, then the sequence of partial sums $\{s_n\}$ is increasing and regular.

Proof

Given a sequence $\{a_k\}$ with non-negative terms, the sequence of partial sums $\{s_n\}$ is increasing since:

$$s_{n+1} = s_n + \underbrace{a_{n+1}}_{\geq 0} \geq s_n, \quad \forall n, \quad \text{hence} \quad \lim_{n \rightarrow +\infty} s_n = \sup_{n \in \mathbb{N}} \{s_n\}$$

by the monotonicity theorem for sequences.

Consequently the sequence $\{s_n\}$ cannot be irregular. It is therefore regular, that is, it either converges or diverges. $\square$

Series with Non-negative Terms

Given a sequence $\{a_k\}$ with non-negative terms:

- ▶ If $\{s_n\}$ is **increasing and bounded**, then it has a finite limit and $\sum a_k$ converges.
- ▶ If $\{s_n\}$ is **increasing and unbounded**, then $\sum a_k$ diverges to $+\infty$.
- ▶ This remark also holds for sequences with eventually non-negative terms, positive terms, or eventually positive terms.

Necessary Condition for Convergence

Theorem 1

If a series $\sum a_k$ is convergent, then $\lim_{k \rightarrow +\infty} a_k = 0$

Proof

By definition of convergent series, the sequence of partial sums $\{s_n\}$ converges to a real number s , i.e., $s_n \rightarrow s \in \mathbb{R}$ as $n \rightarrow +\infty$. Without loss of generality we consider $n_0 = 0$.

We observe that the sequence $\{s_n\}$ can be defined recursively:

$$s_0 = a_0, \quad s_n = s_{n-1} + a_n, \quad \forall n \geq 1$$

Consequently we have:

$$a_n = s_n - s_{n-1}$$

and hence

$$\lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} (s_n - s_{n-1}) = s - s = 0$$



Necessary but Not Sufficient

The convergence of a series therefore implies that $a_k \rightarrow 0$ as $k \rightarrow +\infty$, that is:

$$\sum a_k \text{ convergent} \quad \Rightarrow \quad \lim_{k \rightarrow +\infty} a_k = 0 \quad (1)$$

but the converse is not true (the series $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ of the previous example is a counterexample):

$$\lim_{k \rightarrow +\infty} a_k = 0 \quad \nRightarrow \quad \sum a_k \text{ convergent} \quad (2)$$

Hence the fact that $a_k \rightarrow 0$ as $k \rightarrow +\infty$ is a necessary but not sufficient condition for the convergence of a series.

Contrapositive

- From the contrapositive of (1) we have:

$$\lim_{k \rightarrow +\infty} a_k \neq 0 \Rightarrow \sum a_k \text{ is divergent or irregular}$$

that is, if the limit is not equal to zero, then the series is not convergent.

Summary

Series with non-negative terms:

- ▶ $\{s_n\}$ is increasing: the series converges (if $\{s_n\}$ is bounded) or diverges to $+\infty$
- ▶ This also holds for eventually non-negative terms

Necessary condition:

- ▶ $\sum a_k$ convergent $\Rightarrow a_k \rightarrow 0$
- ▶ The converse is false: $a_k \rightarrow 0$ does not guarantee convergence
- ▶ If $a_k \not\rightarrow 0$ the series does not converge

Tails of Series

Introduction: Tails of Series

- ▶ We see that modifying a finite number of terms **does not change the behavior** of a series
- ▶ We introduce the **tail** of a series: what remains after removing the first terms
- ▶ The tail of a convergent series represents the **error** made when approximating the sum with a partial sum, and it tends to zero

Modifying a Finite Number of Terms

If we modify a finite number of terms of a series, the value of the sum may change, but the behavior of the series remains unchanged.

- ▶ If we modify the value of one term, for example a_{n_0} , for $n \geq n_0$ the sequences of partial sums, the original and the modified one, differ only by that term: hence both converge, both diverge, or both are irregular. The same holds if we modify a finite number of terms.

The results on the behavior of series therefore also hold if the hypotheses are satisfied “eventually”, i.e., from a certain index n_0 onward.

Tail of a Series

Definition 4: of tail of a series

*Given a series $\sum_{k=0}^{\infty} a_k$ and a value $m \in \mathbb{N}$, the series $\sum_{k=m+1}^{\infty} a_k$ is called a **tail of the series**.*

Tail of a Series

- Given a series $\sum_{k=0}^{\infty} a_k$ and $m \in \mathbb{N}$ we have:

$$\sum_{k=m+1}^n a_k = \left(\sum_{k=0}^n a_k \right) - \left(\sum_{k=0}^m a_k \right) = s_n - s_m, \quad \forall n > m \quad (3)$$

hence for every tail (that is, for every m) the associated sequence of partial sums is equal to that of the original series minus a constant.

Every tail of a series has the same behavior as the original series.

Tail as an Error

- Given a convergent series, taking the limit as $n \rightarrow +\infty$ in (3) we obtain the relation:

$$\sum_{k=m+1}^{\infty} a_k = \left(\sum_{k=0}^{\infty} a_k \right) - \left(\sum_{k=0}^m a_k \right) = s - s_m \quad (4)$$

Given $m \in \mathbb{N}$, every tail $\sum_{k=m+1}^{\infty} a_k$ of a convergent series can be interpreted as the **error** that we make when approximating the sum s with the partial sum s_m .

The Tail Tends to Zero

Theorem 2

If a series $\sum a_k$ is convergent, then

$$\underbrace{s - s_m}_{\infty} \rightarrow 0 \quad \text{as } m \rightarrow +\infty$$
$$= \sum_{k=m+1}^{\infty} a_k$$

Proof (1/2)

For every $n > m$, the n -th partial sum of the series is:

$$s_n = s_m + \sum_{k=m+1}^n a_k$$

Taking the limit as $n \rightarrow +\infty$, we have:

$$\begin{aligned} s &= s_m + \lim_{n \rightarrow +\infty} \sum_{k=m+1}^n a_k \\ &= \underbrace{\lim_{n \rightarrow +\infty} \sum_{k=m+1}^n a_k}_{\infty} \\ &= \sum_{k=m+1}^{\infty} a_k \end{aligned}$$

Proof (2/2)

Now taking the limit as $m \rightarrow +\infty$, we have:

$$s = s + \lim_{m \rightarrow +\infty} \sum_{k=m+1}^{\infty} a_k \Rightarrow \lim_{m \rightarrow +\infty} \sum_{k=m+1}^{\infty} a_k = 0$$

that is, the claim of the theorem. □

Interpretation

- ▶ This theorem tells us that, if a series is convergent, then the value $s - s_m$, that is, the error that we make when approximating the sum s with the partial sum s_m , tends to zero as $m \rightarrow +\infty$.
- ▶ In other words, the tail of a convergent series tends to zero as m tends to infinity.

Summary

Tails of series:

- ▶ Modifying a finite number of terms may change the sum, but not the behavior
- ▶ The hypotheses on the behavior may hold only “eventually”
- ▶ The tail $\sum_{k=m+1}^{\infty} a_k$ has the same behavior as the series

Convergent series:

- ▶ $\sum_{k=m+1}^{\infty} a_k = s - s_m$ is the error made when approximating s with s_m
- ▶ This error tends to zero as $m \rightarrow +\infty$

Harmonic Series

Introduction: Harmonic Series

- ▶ The **harmonic series** $\sum_{k=1}^{\infty} \frac{1}{k}$ is one of the most important series
- ▶ We will prove that it **diverges** to $+\infty$, even though its terms tend to zero
- ▶ It is therefore the classical counterexample showing that $a_k \rightarrow 0$ is not enough for convergence

Harmonic Series

Definition 5: of harmonic series

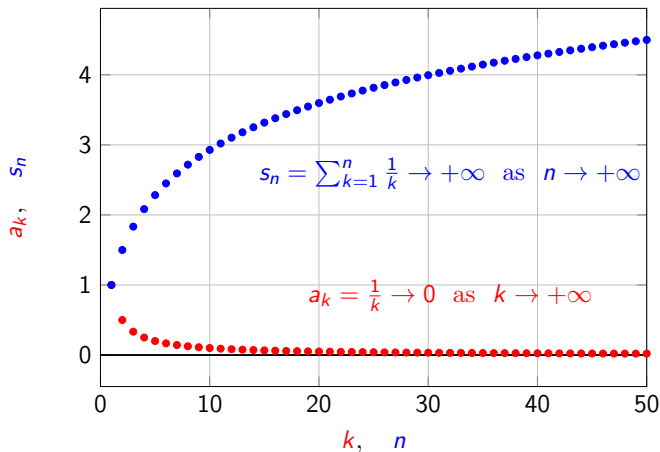
*The **harmonic series** is the series $\sum_{k=1}^{\infty} \frac{1}{k}$*

Theorem 3: behavior of the harmonic series

The harmonic series diverges to $+\infty$

Harmonic Series

- Graphically we have:



A Counterexample

- ▶ The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ is a counterexample that proves (2), since:

$$\lim_{k \rightarrow +\infty} \frac{1}{k} = 0 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges to } +\infty$$

Partial Sums with Index 2^n (1/2)

- Before proving the theorem, we observe that for certain values of n the partial sums are:

$$\underbrace{s_1}_{=s_2^0} = 1 + \frac{0}{2}, \quad \underbrace{s_2}_{=s_2^1} = s_1 + \frac{1}{2} = 1 + \frac{0}{2} + \frac{1}{2} = 1 + \frac{1}{2}$$

$$\underbrace{s_4}_{=s_2^2} = s_2 + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\geq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}} \geq 1 + \frac{1}{2} + \frac{1}{2} = 1 + \frac{2}{2}$$

Partial Sums with Index 2^n (2/2)

$$\underbrace{s_8}_{=s_2^3} = s_4 + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right)}_{\geq \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}} \geq 1 + \frac{2}{2} + \frac{1}{2} = 1 + \frac{3}{2}$$

$$\underbrace{s_{16}}_{=s_2^4} = s_8 + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16} \right)}_{\geq \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{1}{2}} \geq 1 + \frac{3}{2} + \frac{1}{2} = 1 + \frac{4}{2}$$

Proof (1/3)

We prove by induction on 2^n that

$$s_{2^n} \geq 1 + \frac{n}{2}, \quad \forall n \in \mathbb{N}$$

Base case. Let $n = 0$. Then the statement becomes $s_{2^0} \geq 1 + \frac{0}{2}$, i.e., $1 \geq 1$, which is clearly true.

Inductive step. Assume that it is true for 2^{n-1} and let us prove it for 2^n . By the inductive hypothesis, we have:

$$s_{2^{n-1}} \geq 1 + \frac{n-1}{2}$$

Proof (2/3)

Moreover, we have:

$$\begin{aligned} s_{2^n} &= s_{2^{n-1}} + \underbrace{\sum_{k=2^{n-1}+1}^{2^n} \left(\frac{1}{k}\right)}_{\geq 2^{n-1} \cdot \frac{1}{2^n} = \frac{1}{2}} \\ &\geq 2^{n-1} \cdot \frac{1}{2^n} = \frac{1}{2} \end{aligned}$$

hence

$$s_{2^n} \geq 1 + \frac{n-1}{2} + \frac{1}{2} = 1 + \frac{n}{2}$$

which is exactly the desired statement for 2^n .

Proof (3/3)

We therefore have:

$$s_{2^n} \geq 1 + \frac{n}{2} \rightarrow +\infty \quad \text{as } n \rightarrow +\infty$$

and $\{s_n\}$ is unbounded above and also monotonically increasing, since $a_k = 1/k > 0, \forall k \geq 1$. Consequently $\{s_n\}$ diverges to $+\infty$ by the monotonicity theorem for sequences, and the harmonic series diverges to $+\infty$. □

Final Remark

- ▶ The sequence $a_k = 1/k, \forall k \geq 1$, of the harmonic series has positive terms, hence the sequence of partial sums $\{s_n\}$ is increasing, and it is divergent since it is unbounded.

Summary

Harmonic series:

- ▶ $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges to $+\infty$
- ▶ By induction: $s_{2^n} \geq 1 + \frac{n}{2}$, hence $\{s_n\}$ is increasing and unbounded

Counterexample:

- ▶ $\frac{1}{k} \rightarrow 0$ but the series diverges: the condition $a_k \rightarrow 0$ is necessary but not sufficient

Geometric Series

Introduction: Geometric Series

- ▶ The **geometric series** $\sum_{k=0}^{\infty} q^k$ is one of the few series whose sum is known explicitly
- ▶ Its behavior depends only on the **ratio** q : it converges if $|q| < 1$, diverges if $q \geq 1$, is irregular if $q \leq -1$
- ▶ It is a reference model that we will often use to compare other series

Geometric Series

Definition 6: of geometric series

Given $q \in \mathbb{R}$, the *geometric series with ratio q* is the series $\sum_{k=0}^{\infty} q^k$

Theorem 4: behavior and sum of the geometric series

Given the ratio $q \in \mathbb{R}$, the geometric series

$$\sum_{k=0}^{\infty} q^k \text{ is } \begin{cases} \text{convergent} & \text{if } |q| < 1 \\ \text{divergent to } +\infty & \text{if } q \geq 1 \\ \text{irregular} & \text{if } q \leq -1 \end{cases}$$

If the geometric series is convergent, its sum s is $\frac{1}{1-q}$

Proof (1/2)

Given $q \in \mathbb{R}$ and $n \in \mathbb{N}$, the n -th partial sum of the geometric series is:

$$\sum_{k=0}^n q^k = \begin{cases} \frac{1 - q^{n+1}}{1 - q} & \text{if } q \neq 1 \\ n + 1 & \text{if } q = 1 \end{cases}$$

since it equals the sum of the first $n + 1$ terms of the geometric progression. Moreover, we have:

$$\lim_{n \rightarrow +\infty} q^n = \begin{cases} +\infty & \text{if } q > 1 \\ 1 & \text{if } q = 1 \\ 0 & \text{if } |q| < 1 \\ \text{does not exist} & \text{if } q \leq -1 \end{cases}$$

Proof (2/2)

Hence, if $q \neq 1$:

$$\begin{aligned}\lim_{n \rightarrow +\infty} s_n &= \lim_{n \rightarrow +\infty} \frac{1 - q^{n+1}}{1 - q} = \frac{1}{1 - q} \lim_{n \rightarrow +\infty} (1 - q^{n+1}) \\ &= \begin{cases} \frac{1}{1 - q} & \text{if } |q| < 1 \\ +\infty & \text{if } q > 1 \\ \text{does not exist} & \text{if } q \leq -1 \end{cases}\end{aligned}$$

and if $q = 1$ we have:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (n + 1) = +\infty$$



Extension to the case $a \in \mathbb{R}$ (1/2)

- Given the ratio $q \in \mathbb{R}$ and a first term $a \in \mathbb{R}$, the result extends to the series of the terms of the geometric progression:

$$\sum_{k=0}^{\infty} a q^k \text{ is } \begin{cases} \text{convergent} & \text{if } a = 0 \text{ or } |q| < 1 \\ \text{divergent to } +\infty & \text{if } a > 0 \text{ and } q \geq 1 \\ \text{divergent to } -\infty & \text{if } a < 0 \text{ and } q \geq 1 \\ \text{irregular} & \text{if } a \neq 0 \text{ and } q \leq -1 \end{cases}$$

- When it is convergent, its sum is $\frac{a}{1-q}$ if $|q| < 1$, and 0 if $a = 0$

Extension to the case $a \in \mathbb{R}$ (2/2)

- ▶ If $a = 0$ all the terms are zero and therefore $s_n = 0$ for every n
- ▶ If $a \neq 0$, by the product by a constant of summations:

$$s_n = \sum_{k=0}^n a q^k = a \sum_{k=0}^n q^k$$

- ▶ Therefore, by the theorem on the limit of the product of a constant and a sequence, $\{s_n\}$ has the same behavior as the geometric series, with the limit multiplied by a (in the divergent cases the sign of a decides between $+\infty$ and $-\infty$)

Example (1/4)

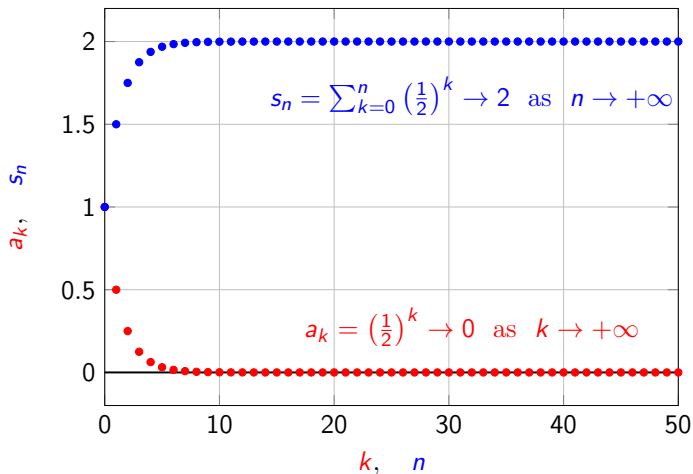
Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \frac{1}{2^k} = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k$$

It is a geometric series with ratio $q = 1/2$, hence it is convergent and its sum s is

$$\frac{1}{1 - 1/2} = 2$$

Example (2/4)



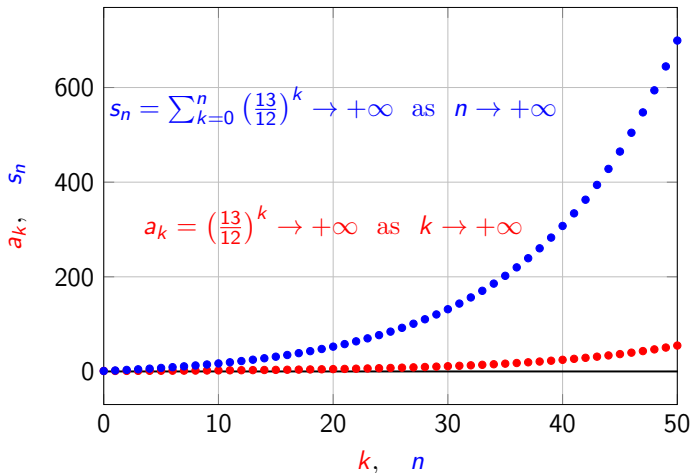
Example (3/4)

Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \left(\frac{13}{12}\right)^k$$

It is a geometric series with ratio $q = 13/12$, hence it is divergent.

Example (4/4)



Summary

Geometric series $\sum_{k=0}^{\infty} q^k$:

- ▶ Converges if $|q| < 1$, with sum $\frac{1}{1-q}$
- ▶ Diverges to $+\infty$ if $q \geq 1$
- ▶ Is irregular if $q \leq -1$

Idea of the proof:

- ▶ For $q \neq 1$: $s_n = \frac{1-q^{n+1}}{1-q}$, and we study the limit of q^{n+1}

Generic first term $a \sum_{k=0}^{\infty} a q^k$:

- ▶ Same behavior, with sum $\frac{a}{1-q}$ if $|q| < 1$; in the divergent cases the sign of a decides between $+\infty$ and $-\infty$

Arithmetic series

Introduction: Arithmetic series

- ▶ The **arithmetic series** $\sum_{k=0}^{\infty} (a + k d)$ adds up the terms of an arithmetic progression
- ▶ It is the natural companion of the geometric series: there the terms are multiplied by q , here d is added each time
- ▶ We will see that, unlike the geometric one, the arithmetic series never converges (except in the trivial case) and is never irregular

Arithmetic series

Definition 7: of arithmetic series

*Given the first term $a \in \mathbb{R}$ and the common difference $d \in \mathbb{R}$, the **arithmetic series** is the series $\sum_{k=0}^{\infty} (a + k d)$*

The terms of the arithmetic series are the terms of the arithmetic progression with first term a and common difference d , that is, $a, a + d, a + 2d, \dots$. As for the geometric series, the first term is in position $k = 0$.

Behavior of the arithmetic series

Theorem 5: behavior and sum of the arithmetic series

Given the first term $a \in \mathbb{R}$ and the common difference $d \in \mathbb{R}$, the arithmetic series

$$\sum_{k=0}^{\infty} (a + k d) \text{ is } \begin{cases} \text{divergent to } +\infty & \text{if } d > 0 \\ \text{divergent to } -\infty & \text{if } d < 0 \\ \text{divergent to } +\infty & \text{if } d = 0 \text{ and } a > 0 \\ \text{divergent to } -\infty & \text{if } d = 0 \text{ and } a < 0 \\ \text{convergent} & \text{if } d = 0 \text{ and } a = 0 \end{cases}$$

If the arithmetic series is convergent, its sum s is 0

Proof (1/2)

Given $a, d \in \mathbb{R}$ and $n \in \mathbb{N}$, the n -th partial sum of the arithmetic series is:

$$\begin{aligned} s_n &= \sum_{k=0}^n (a + k d) = a(n+1) + d \left(\frac{n(n+1)}{2} \right) \\ &= \frac{d}{2} n^2 + \left(a + \frac{d}{2} \right) n + a \end{aligned}$$

since it equals the sum of the first $n+1$ terms of the arithmetic progression.

Therefore, if $d \neq 0$, collecting the term of highest degree:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \frac{d}{2} n^2 \left(1 + \underbrace{\frac{2a+d}{dn}}_{\rightarrow 0} + \underbrace{\frac{2a}{dn^2}}_{\rightarrow 0} \right) = \begin{cases} +\infty & \text{if } d > 0 \\ -\infty & \text{if } d < 0 \end{cases}$$

Proof (2/2)

And if $d = 0$ we have $s_n = a(n + 1)$ and therefore:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} a(n + 1) = \begin{cases} +\infty & \text{if } a > 0 \\ -\infty & \text{if } a < 0 \\ 0 & \text{if } a = 0 \end{cases}$$



Only a trivial case of convergence

- ▶ The arithmetic series is convergent only in the trivial case $a = d = 0$, in which all its terms are zero, and it is never irregular (unlike the geometric series, which is irregular for $q \leq -1$)
- ▶ This also follows from the necessary condition for convergence: the general term is $a_k = a + k d$ and

$$\lim_{k \rightarrow +\infty} (a + k d) = \begin{cases} +\infty & \text{if } d > 0 \\ -\infty & \text{if } d < 0 \\ a & \text{if } d = 0 \end{cases}$$

that is, $a_k \rightarrow 0$ only if $d = 0$ and $a = 0$

- ▶ The necessary condition does not say, however, *which* the behavior is: it is the computation of the partial sums that establishes that the series is divergent, and with which sign

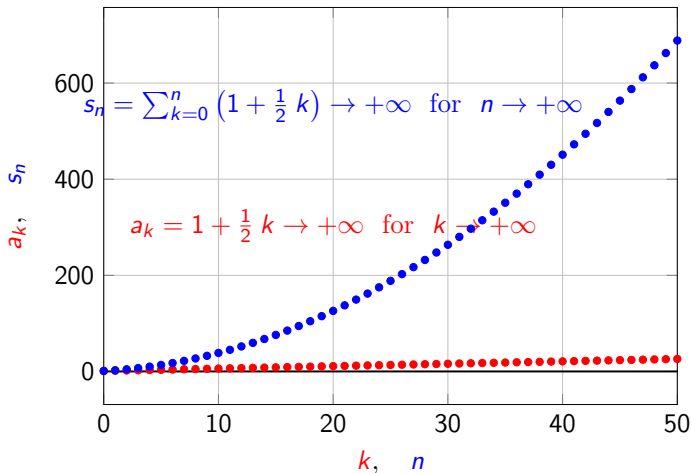
Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \left(1 + \frac{1}{2} k \right)$$

It is an arithmetic series with first term $a = 1$ and common difference $d = 1/2$. Since $d > 0$, the series is divergent to $+\infty$.

Example (2/2)



Summary

Arithmetic series $\sum_{k=0}^{\infty} (a + k d)$:

- ▶ Diverges to $+\infty$ if $d > 0$, to $-\infty$ if $d < 0$
- ▶ If $d = 0$ it diverges to $+\infty$ or $-\infty$ according to the sign of a
- ▶ It converges only if $a = d = 0$, with sum 0; it is never irregular

Idea of the proof:

- ▶ $s_n = \frac{d}{2} n^2 + \left(a + \frac{d}{2}\right) n + a$, that is, a polynomial of degree two in n

Telescoping Series



Introduction: Telescoping Series

- ▶ A **telescoping series** has terms of the form $b_k - b_{k+1}$: in the partial sums almost all terms cancel out
- ▶ Its behavior and its sum depend only on the limit of the sequence $\{b_k\}$
- ▶ We will see **Mengoli's series** and how to use partial fraction decomposition to recognize a telescoping series

Telescoping Series

Definition 8: of telescoping series

A *telescoping series* is a series of the form:

$$\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) \quad \text{where } \{b_k\} \text{ is a sequence}$$

Behavior of the Telescoping Series

Theorem 6: behavior and sum of the telescoping series

A telescoping series converges, diverges or is irregular according to whether the sequence $\{b_k\}$ converges, diverges or is irregular, respectively. If $\lim_{k \rightarrow +\infty} b_k = \ell \in \mathbb{R}^$ then*

$$\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) = b_{n_0} - \underbrace{\lim_{k \rightarrow +\infty} b_k}_{=\ell \in \mathbb{R}^*}$$

Moreover, if $\ell \in \mathbb{R}$, then the telescoping series is convergent and its sum s is $b_{n_0} - \ell$

Proof

We have

$$\begin{aligned}s_n &= \sum_{k=n_0}^{n_0+n} (b_k - b_{k+1}) \\&= (b_{n_0} - b_{n_0+1}) + (b_{n_0+1} - b_{n_0+2}) + \dots + (b_{n_0+n} - b_{n_0+n+1}) \\&= b_{n_0} - b_{n_0+n+1}\end{aligned}$$

hence:

$$\begin{aligned}\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) &= \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (b_{n_0} - b_{n_0+n+1}) \\&= b_{n_0} - \lim_{n \rightarrow +\infty} b_n\end{aligned}$$



Mengoli's Series

Observation 2

*The series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$, called **Mengoli's series**, is convergent and its sum s is equal to 1.*

Proof

We have:

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{(k+1) - k}{k(k+1)} = \sum_{k=1}^{\infty} \underbrace{\frac{1}{k}}_{=b_k} - \underbrace{\frac{1}{k+1}}_{=b_{k+1}}$$

Mengoli's series therefore has the form:

$$b_k - b_{k+1} \quad \text{with} \quad b_k = \frac{1}{k}$$

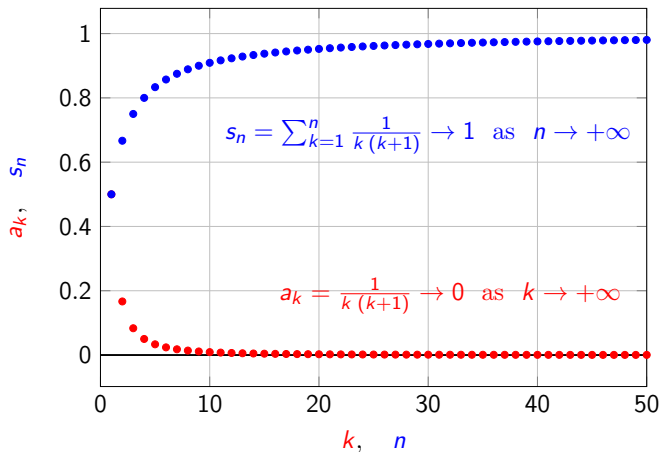
and moreover

$$\lim_{k \rightarrow +\infty} b_k = \lim_{k \rightarrow +\infty} \frac{1}{k} = 0$$

hence it is a convergent telescoping series ($n_0 = 1$) and moreover $s = b_1 = 1$. □

Mengoli's Series

► Graphically we have:



Alternative Proof

For every $n \geq 1$, the value of the partial sum of Mengoli's series is:

$$\begin{aligned} s_n &= \sum_{k=1}^n \frac{1}{k(k+1)} = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1} \end{aligned}$$

Hence we have

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n+1} \right) = 1$$

Consequently Mengoli's series is convergent and its sum s is 1. $\square$

Example (1/4)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)}$$

We have:

$$\begin{aligned} \frac{1}{(3k+2)(3k+5)} &= \frac{\alpha}{3k+2} - \frac{\beta}{3k+5} \\ &= \frac{\alpha(3k+5) - \beta(3k+2)}{(3k+2)(3k+5)} \\ &= \frac{3(\alpha - \beta)k + 5\alpha - 2\beta}{(3k+2)(3k+5)} \end{aligned}$$

Example (2/4)

Equating the coefficients of the numerators:

$$\Rightarrow \begin{cases} 3(\alpha - \beta) = 0 \\ 5\alpha - 2\beta = 1 \end{cases} \Rightarrow \alpha = \beta, \quad 3\alpha = 1 \quad \text{hence} \quad \alpha = \beta = \frac{1}{3}$$

Example (3/4)

Consequently:

$$\begin{aligned}\frac{1}{(3k+2)(3k+5)} &= \frac{1}{3} \left(\frac{1}{3k+2} - \frac{1}{3k+5} \right) \\ &= \frac{1}{3} \left(\underbrace{\frac{1}{3k+2}}_{=b_k} - \underbrace{\frac{1}{3(k+1)+2}}_{=b_{k+1}} \right)\end{aligned}$$

The series therefore has the form:

$$b_k - b_{k+1} \quad \text{with} \quad b_k = \frac{1}{3k+2}$$

and moreover

$$\lim_{k \rightarrow +\infty} b_k = \lim_{k \rightarrow +\infty} \frac{1}{3k+2} = 0$$

Example (4/4)

It is therefore a telescoping series ($n_0 = 1$), and we have:

$$\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)} = \frac{1}{3} \left(b_1 - \underbrace{\lim_{k \rightarrow +\infty} b_k}_{\rightarrow 0} \right) = \frac{1}{3} b_1 = \frac{1}{15}$$

Hence it is a convergent telescoping series and its sum s is $\frac{1}{15}$.

Summary

Telescoping series $\sum_{k=n_0}^{\infty} (b_k - b_{k+1})$:

- ▶ $s_n = b_{n_0} - b_{n_0+n+1}$: it has the same behavior as $\{b_k\}$
- ▶ If $b_k \rightarrow \ell \in \mathbb{R}$, it converges with sum $b_{n_0} - \ell$

Examples:

- ▶ Mengoli's series: $\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = 1$
- ▶ With partial fractions: $\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)} = \frac{1}{15}$

Summary (1/2)

Definitions:

- ▶ Series $\sum a_k$, partial sums s_n , behavior and sum of the series
- ▶ Tail of a series: same behavior as the series, and it is the error $s - s_m$

Properties:

- ▶ Non-negative terms: the series converges or diverges to $+\infty$
- ▶ $\sum a_k$ convergent $\Rightarrow a_k \rightarrow 0$ (necessary, not sufficient condition)
- ▶ Modifying a finite number of terms does not change the behavior

Summary (2/2)

Notable series:

- ▶ **Harmonic** $\sum \frac{1}{k}$: diverges to $+\infty$, even though $\frac{1}{k} \rightarrow 0$
- ▶ **Geometric** $\sum a q^k$: converges to $\frac{a}{1-q}$ if $|q| < 1$, diverges if $q \geq 1$, irregular if $q \leq -1$
- ▶ **Arithmetic** $\sum (a + k d)$: always diverges, except in the trivial case $a = d = 0$
- ▶ **Telescoping** $\sum (b_k - b_{k+1})$: sum $b_{n_0} - \lim b_k$
- ▶ **Mengoli** $\sum \frac{1}{k(k+1)} = 1$